

UAV Detection using micro-Doppler Spectrograms and Deep Learning

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Abstract—Commercial UAVs have become increasingly relevant in recent years, as their usage has proliferated in a variety of areas. The ability to detect them is of considerable importance. Current deep learning approaches use proprietary datasets due to the small size of public datasets online. In an effort to mitigate that, my project utilizes transfer learning by using CNN ResNet18, data augmentation, and pretraining on synthetic radar data using RadarSimPy. To deal with the heavy class imbalance in existing datasets, focal loss and augmentation was used. Compared to a ResNet18 CNN fine-tuned on just the dataset, the model using synthetic data and augmented data showed small but significant increases in performance, with the F1 score increasing from 40.2% to 52.77%.

Index Terms—Classification, Transfer Learning, mm-Wave Radar, Spectrograms, Machine Learning

I. INTRODUCTION

Commercial Unmanned Aerial Vehicles, or UAVs, are small unmanned aircraft that have seen increased usage in recent times due to their declining costs and improvements in technology. Use cases range from relatively benign usages, like surveillance, to much more grave use cases, like the usage of FPV drones in modern warfare against vehicles and personnel [3]. Consequently, the ability to successfully detect these vehicles is of significant importance. Current Deep Learning methods of detection primarily rely on visual data, which struggles with poor visibility environments (fog, rain, low-light conditions) and clutter.

Radar performs better in those situations due to it measuring the return of electromagnetic energy. In the case of small commercial drones, frequencies in the mmWave are best suited to this task as larger frequencies fail to generate a return due to the size of the aircraft. The primary data type that will be analyzed are micro-Doppler spectrograms. Essentially, the Doppler signature of an object is measured over a time period, which generates a 2D mapping that can then be fed into a CNN. Doppler signatures are used to measure the velocity of the object through the Doppler Effect. Micro-Doppler spectrograms are particularly useful classifying UAVs as they generate distinct Doppler signatures primarily due to the high RPM of their propellers which differ significantly from spectrograms from other objects like a walking human. However, a significant issue in applying deep learning to radar based approaches of classification is the limited number of datasets that are publicly available, due in part to the high cost

of equipment to create data. For example, the dataset used in this paper only yielded 350 samples, 50 of which were of UAVs [1]. A possible approach to help reduce the effect of this problem would be to pretrain a model on a synthetic dataset of radar data, so that the samples in the dataset can then be fine-tuned on this model. The goal of this project is to develop a CNN model capable of detecting and classifying UAVs within a short period of time based on their mmWave radar signature. A secondary goal is to create a physics-based radar sample simulator to supplement existing datasets. What is outside the scope of this project and will not be addressed is the development of a physical system in which the model can be deployed.

II. METHOD

I evaluated and compared the performance of ResNet18 given a model that was pretrained on synthetic data before training on the dataset and a model that trained on the dataset only. Both models utilized transfer learning from the existing weights "IMAGENET1K_V1". Additive augmentation was also used to help the model reach convergence more quickly. Fig.1 shows the pipeline from generating synthetic data, pre-processing the dataset, splitting the data into a training set and test set, performing 5-fold cross validation on the training set, and then testing and aggregating results.

A. Dataset

I used the publicly available Open Radar Dataset [1]. The dataset is a python dictionary that primarily contains doppler spectra, with each dictionary entry being a radar signature of an entire track. Each track is around 30 ms long, with 350 signatures available. There are 50 full track signatures for UAVs, with the rest of the dataset being signatures of moving bicycles, humans, and motor vehicles. The labels in the model are "UAV" and "non-UAV".

B. Additive Augmentations

Due to the class imbalance with UAVs representing the minority class, additive augmentation was applied to UAV samples. Types of augmentations included amplitude scaling, adding noise, doppler shifts, vertical flips, and horizontal flips. This increased the number of UAV samples from an original of 50 to a total of 250.

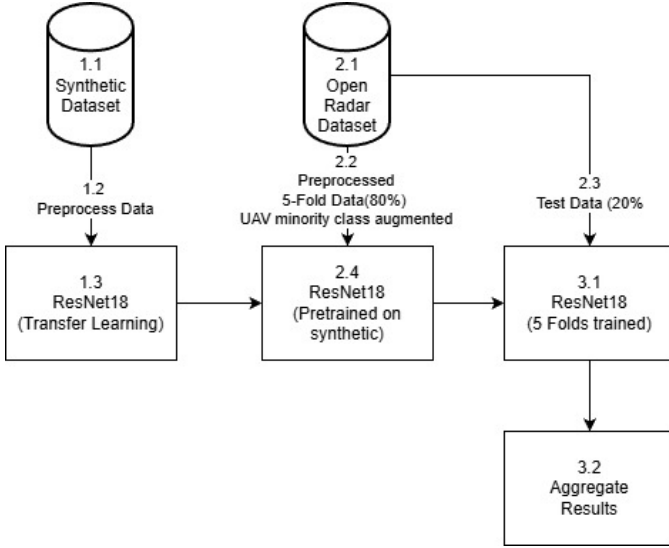


Fig. 1. Overall architecture of the synthetic data approach for classifying UAVs. 1) Synthetic dataset is generated, preprocessed, and pretrains a ResNet18 network that has its pretrained weights. 2) Actual dataset is split 80/20, with 80% of data being used in a 5-Fold cross-validation split, and 20% used as a final test set. 3) Trained models for each fold are tested and then aggregated.

C. Preprocessing

Raw IQ data is loaded from the dataset, a range bin is selected, then a Short Time Fourier Transform (STFT) is applied to convert the 1D signal 2D spectrogram. The spectrogram is then converted to magnitude of return, which is then normalized by scaling it in terms of decibels and applying zero mean and unit variance. Then, a channel dimension is added in order to fit the dimensions of the ResNet. After converting the spectrogram to a PyTorch tensor the data is ready to be fed into the model.

D. Synthetic Data

Synthetic data was generated with RadarSimPy, a python based library for modelling and simulating radar systems [2]. 5000 samples were generated, with 2000 UAV samples, 1000 vehicle samples, 1000 bicycle samples, and 1000 pedestrian samples. This synthetic data was then used to pretrain the ResNet18 model before training with actual data.

E. Transfer Learning and Training

ResNet18 with existing "IMAGENET1K_V1" weights was main architecture used. Due to the class imbalance, focal loss was also applied. The model was trained for 30 epochs, with the epoch that had the lowest loss on the validation set being selected as the best model for the fold.

III. EXPERIMENTS AND RESULTS

Accuracy, precision, recall, specificity and F1 were used to evaluate the overall performance of the models. The F1 score will be used as the most important metric as it takes into account both precision and recall.

TABLE I
RESULTS (ACCURACY AND F1)

Model Type	Accuracy	F1 Score
ResNet18	81.70%	40.2%
Pretrained ResNet18	82.57%	37.20%
Augmented ResNet18	78.00%	41.3%
Augmented + Pre-trained ResNet18	78.86%	52.77%

TABLE II
RESULTS (PRECISION, RECALL, AND SPECIFICITY)

Model Type	Precision	Recall	Specificity
ResNet18	38.2%	44.00%	88.00%
Pretrained ResNet18	39.80%	39.00%	90.00%
Augmented ResNet18	37.90%	62.00%	80.70%
Augmented + Pre-trained ResNet18	39.71%	80.00%	78.67%

A. Comparison of Models on Original vs Augmented Dataset

The Augmented Dataset significantly increased model recall at the cost of slightly hurting other metrics. For example, on the models without pretraining on the synthetic dataset, accuracy decreased by 3.7%, F1 Scores went up 1.1%, precision went down by 0.3%, and recall went up by 18%. Augmentation, as expected, helps boost the model's ability to recognize UAVs, but also damages its ability to recognize non-UAVs.

B. Comparison of Models on Synthetic Pretraining

Pretraining on a synthetic dataset showed smaller but significant improvements in model performance. For the augmented models, pretraining on a synthetic dataset boosted performance on every metric except specificity. Accuracy went up by 0.86%, F1 by 11.47%, precision by 1.81%, and recall by 18%. The non augmented models showed mixed performance, with F1 decreasing from 40.2% to 37.2%, recall decreasing from 44.00% to 39.00%, accuracy increasing by 0.87%, precision by 1.6%, and specificity by 2%.

IV. CONCLUSION

In this project, data augmentation and synthetic data generation were used in order to increase the efficacy of small radar datasets for UAVs. My findings suggest that these methods are somewhat effective at increasing model performance, with the F1 score increasing from 46.15% to 52.77% after both methods were applied. I conclude that augmenting the dataset was moderately effective at improving model performance by reducing the disparity between the majority and minority class, and that pretraining on synthetic datasets helped the model perform slightly better.

Although model performance increased with augmentation and synthetic data, the most effective way to increase performance is still to increase the amount of data available. This is especially true for small datasets like this one where only 350

samples were used. Furthermore, more accurate proprietary simulators could increase performance even more with more accurate and representative synthetic datasets.

In the future, I will test different architectures using this approach, and use better sources for synthetic data. I will also attempt to generalize to other datasets that use different sensors, as another issue in this area of research is that models don't generalize well across sensors.

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